BIO 101 — Midterm Exam — Fall 2024

Dr. Sarah — Time: 60 minutes — Total: 30 marks

Part I: Choose the correct answer (1 mark each)

1. Which of the following is NOT one of the principles of the cell theory?

   a. All living things are made of cells
   b. The cell is the basic unit of structure and function
   c. Cells arise from pre-existing cells
   d. All cells contain a nucleus

2. Which of the following organelles is the site of protein synthesis?

   a. Golgi apparatus
   b. Ribosomes
   c. Mitochondria
   d. Nucleus

3. The Na^+/K^+ pump moves:

   a. 3 Na^+ in and 2 K^+ out
   b. 2 Na^+ out and 3 K^+ in
   c. 3 Na^+ out and 2 K^+ in
   d. 2 Na^+ in and 2 K^+ out

4. Which of the following best describes enzymes?

   a. They are consumed during the reaction
   b. They increase the activation energy
   c. They lower the activation energy
   d. They are lipids

Part II: True or False (1 mark each)

5. Osmosis is a type of active transport. ( )

6. Mitochondria have their own DNA. ( )

7. Competitive inhibitors bind to the active site. ( )

Part III: Answer the following

8. Define the fluid mosaic model. (3 marks)

9. Compare the lock and key model with the induced fit model. (6 marks)